



PHYSICS CH.3 — FLUID MECHANICS — FULL MIND MAP

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Topics in this chapter:

1. Branches of Fluid Mechanics
2. Archimedes' Principle: Float, Sink & Upthrust
3. Density, Relative Density & Buoyant Force Formula
4. Capillarity, Viscosity & Evaporation
5. Ice-Water Volume Problem & Physical States
6. Real-Life Applications
7. Numerical Formulas: Buoyant Force & Pump Power
8.  Traps / Formula Cheat Sheet (Q381–Q404)

1. Branches of Fluid Mechanics

- Hydrostatics = fluid (gas ya liquid) ke rest (vishram) avastha ke properties ka adhyayan — specially pressure in fluid ya immersed body par fluid dwara lagaya gaya pressure. [Q381]
- Hydrodynamics = fluid ke motion (gati) ka adhyayan. [Q381]
- Optics = light ke behavior aur properties ka adhyayan; Thermodynamics = heat aur energy ke beech relation; Astrophysics = celestial objects ke physical properties. [Q381]

2. Archimedes' Principle: Float, Sink & Upthrust

- Archimedes' Principle: Jab koi body fluid mein partially ya fully immersed hoti hai, toh us par ek upward buoyant force lagta hai jo displaced fluid ke weight ke barabar hota hai. [Q383, Q386, Q391, Q393, Q395, Q401]
- Buoyant force (Upthrust) = $F_b = \rho \times g \times V$, jahan:
 - ρ = fluid ka density, g = acceleration due to gravity, V = displaced volume (object ka submerged volume). [Q387, Q396, Q401]
- Buoyant force ka magnitude fluid ke density aur displaced volume par depend karta hai — direction hamesha upward hoti hai. [Q387]
- Float vs Sink rules:
 - Object floats → jab object ka density liquid se kam ho. [Q385, Q388, Q390, Q400]
 - Object sinks → jab object ka density liquid se zyada ho. [Q385, Q394, Q398]
 - Same density → object liquid mein kahi bhi suspend ho sakta hai (neither float nor sink completely). [Q388]
- Floating object ka apparent weight = Zero hota hai kyunki buoyant force exactly weight ke barabar hoti hai. [Q389]
- Sinking object ke liye buoyant force object ke weight se kam hota hai. [Q394]
- Object 1/3 volume submerged ho to float kare → object ka density = $(1/3) \times$ water density. [Q391]
- Spring balance reading (apparent weight) = Actual weight – Buoyant force = $mg - \rho g V$. [Q397, Q401]
 - Jab object completely submerged ho → maximum fluid displaced → maximum buoyant force → minimum apparent weight. [Q397]
 - Partially submerged ya surface par hone par kam buoyant force → zyada apparent weight. [Q397]

3. Density, Relative Density & Buoyant Force Formula

- Density (ρ) = mass/volume; SI unit = kg/m^3 . [Q385, Q388, Q392]
- Iron nail ka density $\approx 7.8 \text{ g/cm}^3$; water ka density = 1 g/cm^3 (25°C par). [Q398]
- Relative Density = substance ka density / reference substance ka density; unitless hota hai. [Q392]
- Ice ka density water se kam hota hai ($\approx 0.92 \text{ g/cm}^3$) — isliye ice floats; iska unique crystalline structure hai. [Q400]
- ⚠ Trap: Generally solids liquids se zyada dense hote hain, lekin water exception hai — ice (solid) liquid water se kam dense hoti hai. [Q400]

4. Capillarity, Viscosity & Evaporation

- Capillarity = liquids ka narrow spaces/tubes mein gravity ke against rise ya draw hone ka phenomenon. [Q382]
 - Examples: Kerosene oil lantern ki wick mein upar chadhna, blotting paper se ink absorb hona, plants mein water roots se leaves tak jaana. [Q382]
 - NOT capillarity: Osmosis = semipermeable membrane se solvent ka movement; Diffusion = high concentration se low concentration mein molecules ka movement. [Q382]
- Viscosity = fluid ke flow ka resistance; liquid ka "internal friction". [Q384]
 - SI unit = $\text{N}\cdot\text{s/m}^2$ ya Pascal-second ($\text{Pa}\cdot\text{s}$). [Q384]
 - High viscosity examples: Honey, Molasses, Melted chocolate, Engine oil. [Q384]
 - NOT viscosity: Rigidity = solid ka shape na badal paana; Morbidity = disease related; Rancidity = fat/oil ka spoil hona. [Q384]
- Evaporation = liquid ka vapour mein convert hona boiling point se niche ke temperature par. [Q399]
 - Liquid ke surface ke kuch particles mein zyada kinetic energy hoti hai — ye particles attraction force break karke vapour ban jaate hain. Evaporation cooling effect deta hai. [Q399]

5. Ice-Water Volume Problem & Physical States

- Glass completely water aur ice se bhara ho aur ice melt ho jaye $\rightarrow$ water level same rehta hai. [Q393]
 - Reason: Floating ice apne weight ke barabar water displace karti hai; melt hone par bhi wahi volume ka water banegi. [Q393]
- Water ke 3 states:
 - Solid (Ice) $\rightarrow 0^\circ\text{C}$ se niche. [Q402]
 - Liquid (Water) $\rightarrow 0^\circ\text{C}$ se 100°C ke beech. [Q402]
 - Gas (Steam/Vapour) $\rightarrow 100^\circ\text{C}$ se upar. [Q402]
- 12°C par water liquid state mein hota hai. [Q402]
- Water (H_2O) covalent bonds se bana; polar nature; amphoteric; universal solvent (high dielectric constant ki wajah se). [Q402]

6. Real-Life Applications

- Archimedes' Principle par based:
 - Ships & Submarines ki designing. [Q383]
 - Lactometers $\rightarrow$ milk ki purity check karne ke liye. [Q383, Q386]
 - Hydrometers $\rightarrow$ liquid ka density measure karne ke liye; weighted sealed glass bulb hota hai — flotation depth

density batata hai. [Q383, Q386]

– Boat sailing, Iceberg floating, Helium balloon rising. [Q387]

- Bernoulli's Principle par based (NOT Archimedes'):

– Airplane wings ki designing — aerodynamics. [Q383]

– Periscope → Laws of Reflection par based. [Q386]

– Potentiometer → potential difference directly proportional to wire length. [Q386]

7. Numerical Formulas: Buoyant Force & Pump Power

- Buoyant Force numerical mein direct formula:

– $F_b = \text{Weight in air} - \text{Weight in liquid (apparent weight loss)}$. [Q403]

– Example: Air mein 4 N, water mein 2 N → $F_b = 4 - 2 = 2 \text{ N}$. [Q403]

– Yeh Archimedes' principle ka practical application hai — buoyant force utna hi hota hai jitna weight reduce hota hai. [Q403]

- Pump Power numerical:

– Formula: $\text{Power} = \text{Work done} / \text{Time} = (m \times g \times h) / t$. [Q404]

– Example: $\text{Power} = 1 \text{ kW} = 1000 \text{ W}$, $g = 10 \text{ m/s}^2$, $h = 10 \text{ m}$, $t = 60 \text{ s}$. $1000 = (m \times 10 \times 10) / 60 \rightarrow m = (1000 \times 60) / 100 = 600 \text{ kg}$. [Q404]

– Unit check karte raho: Power in Watts, time in seconds, mass in kg, g in m/s^2 , h in meters. [Q404]

8. ⚠ Traps / Formula Cheat Sheet (Q381–Q404)

- ⚠ Trap: Airplane wings Bernoulli's principle par based hain, Archimedes' principle par nahi. [Q383]

- ⚠ Trap: Buoyant force $= \rho \times g \times V$; object ka mass nahi, displaced volume (V) use hota hai. [Q396]

- ⚠ Trap: Object float karega jab uska density liquid se kam ho; sink karega jab zyada ho; same density ho toh suspend. [Q388, Q390]

- ⚠ Trap: Ice melt hone par water level same rehta hai — increase ya decrease nahi hota. [Q393]

- ⚠ Trap: Spring balance ka minimum reading tab aata hai jab object completely submerged ho (max buoyant force). [Q397]

- ⚠ Trap: Apparent weight $= mg - \rho g V$; zero tab hota hai jab object float kare. [Q389, Q401]

- ⚠ Trap: Capillarity $\neq$ Osmosis $\neq$ Diffusion — teeno alag-alag phenomena hain. [Q382]

- ⚠ Trap: Viscosity = liquid ka internal friction; Rigidity = solid ka property. [Q384]

- ⚠ Trap: Ice ka density water se kam hota hai — yeh water ka unique property hai (generally solids liquids se zyada dense hote hain). [Q400]

- ⚠ Trap: Numerical mein time hamesha seconds mein convert karo — 1 minute = 60 s. [Q404]

- ⚠ Trap: Buoyant force = Weight loss (air weight – liquid weight); direct subtraction se nikalta hai. [Q403]